System Modeling

Find and Fix the UML Mistakes

Q1. A university library system has a Book class. Each book has a title, an ISBN, and an availability status. The system can check whether a book is available.

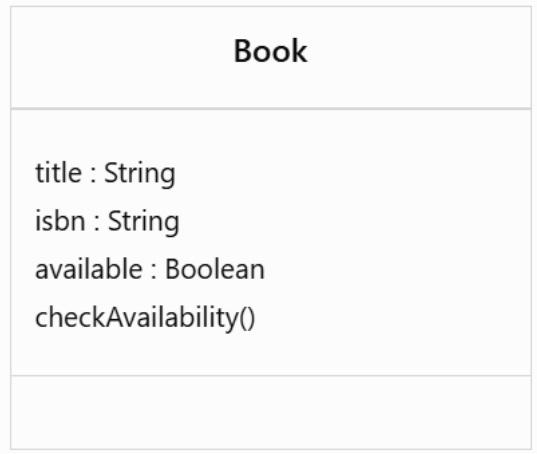
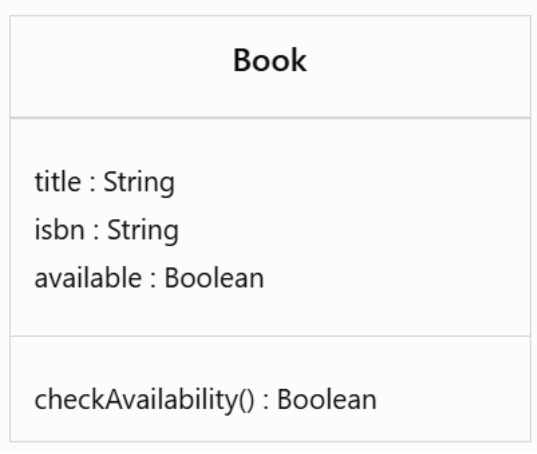


Figure 1. Incorrect Book class diagram

Identify the mistake and redraw the class correctly.

Answer: checkAvailability() is an operation, not an attribute. It belongs in the bottom compartment of the UML class.

Corrected diagram:



Q2. In a university library system:

- **One student can borrow several books over time.**
- **Every loan record belongs to exactly one student.**
- **A student may have no loan records.**

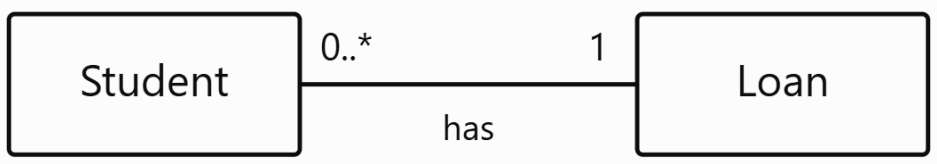


Figure 2. Incorrect multiplicities

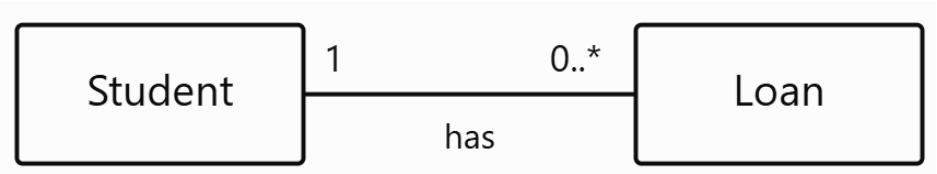
Are the multiplicities correct? If not, identify which values should be changed.

Answer: The multiplicities have been reversed.

Near Student: 1 (each Loan belongs to exactly one Student)

Near Loan: 0..* (A Student may have zero or many Loans)

Corrected diagram:



Q3. A university system has a general class called Person. There are two specialised classes: Student and Teacher. Both inherit the common attributes of Person.

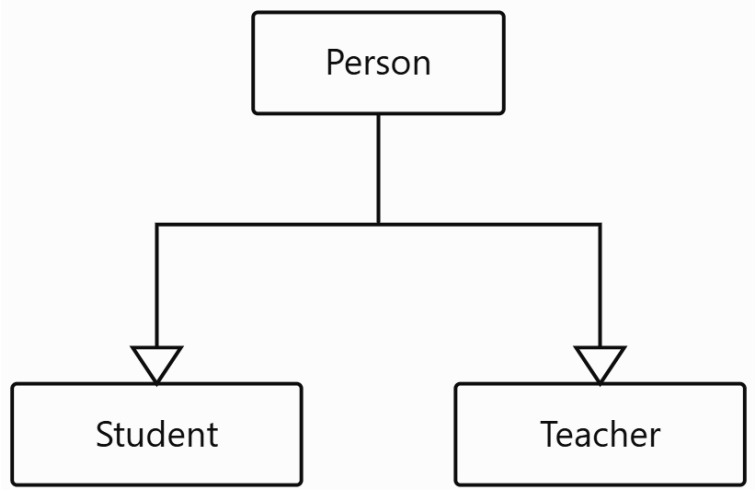
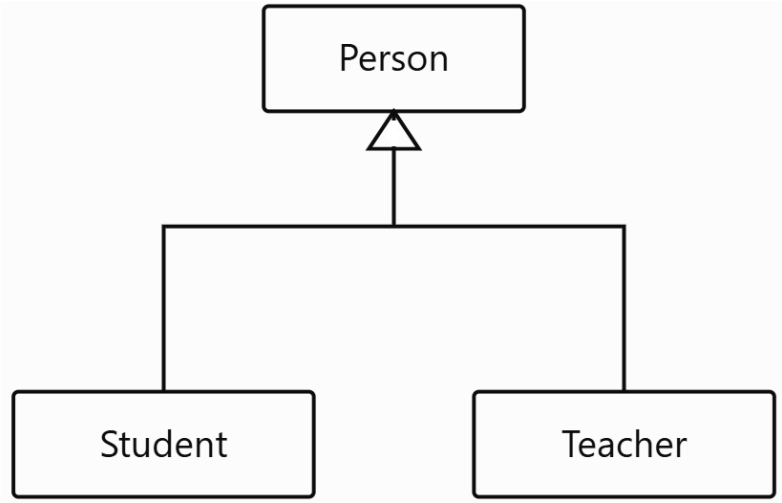


Figure 3. Incorrect generalisation

Find the error in the inheritance relationship.

Answer: The hollow triangular arrowheads point toward the subclasses (Student and Teacher). They should point toward the superclass (Person).

Corrected diagram:



Q4. A hospital system maintains a PatientRecord that contains multiple Consultation records. The PatientRecord represents the whole, while the consultations are its parts.

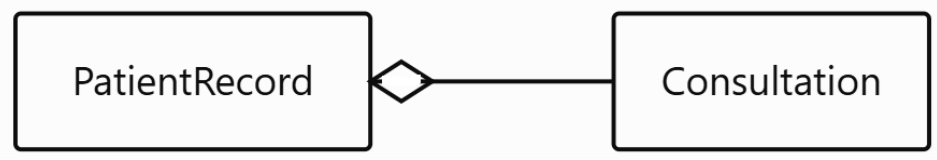


Figure 4. Incorrect aggregation

What is wrong with the aggregation relationship?

Answer: The hollow diamond is beside Consultation, but it should be beside PatientRecord, because PatientRecord represents the whole.

Corrected diagram:



Q5. A student accesses a university system to view examination results. The intended sequence is:

1. **The student requests the results.**
2. **The system verifies the student's login status.**
3. **If the student is authenticated, the system retrieves the results.**
4. **The system displays the results.**

However, an engineer has produced the following sequence diagram.

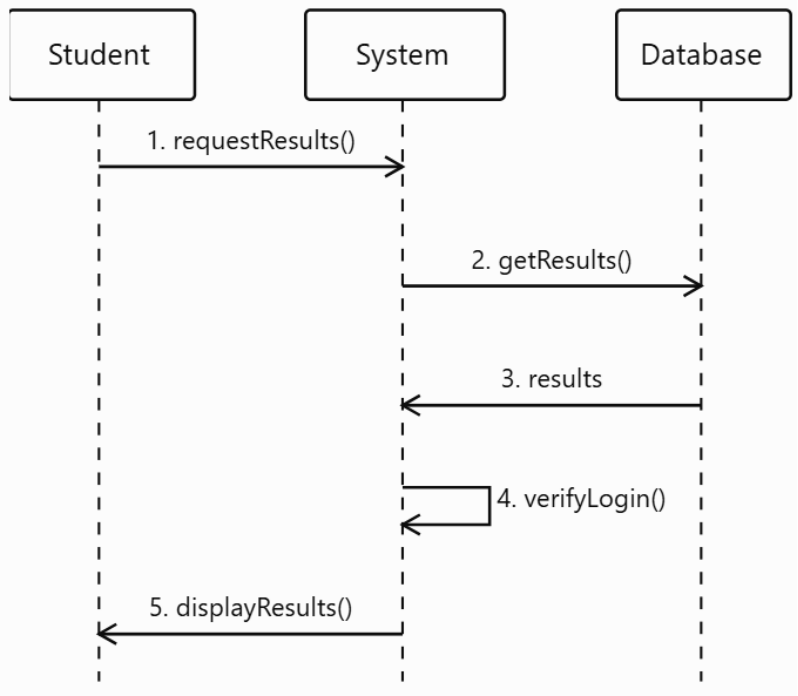


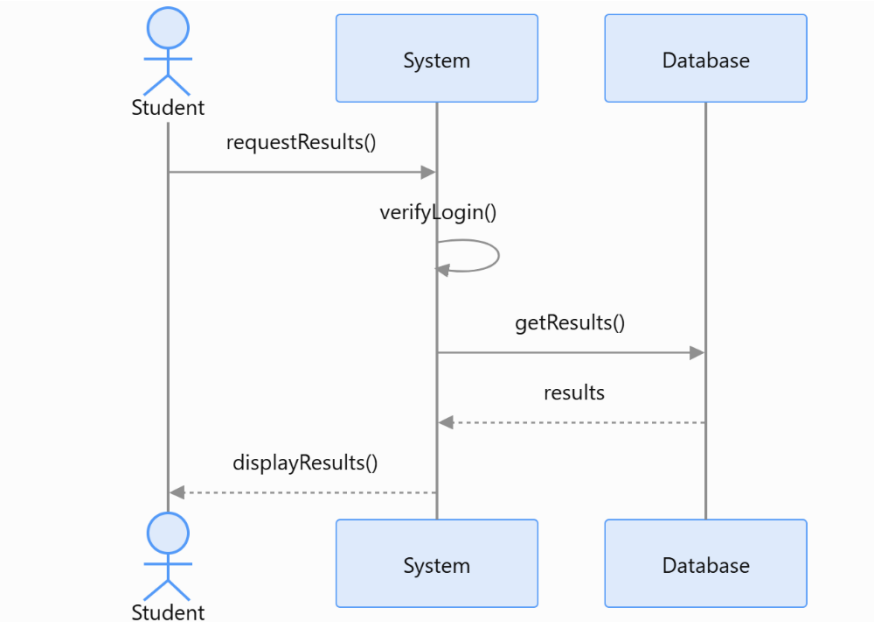
Figure 5. Incorrect sequence diagram

Identify the mistake in the order of operations. Explain the problem and specify the correct message order.

Answer: The system retrieves the examination results before verifying the student's login status. This violates the requirements and may expose information to unauthenticated users.

The system must authenticate the student before retrieving and displaying the results.

Corrected sequence diagram:



Q6. In a library management system, a student can search for books and borrow books. The system stores book information in its internal database. The use diagram of the system is shown below.

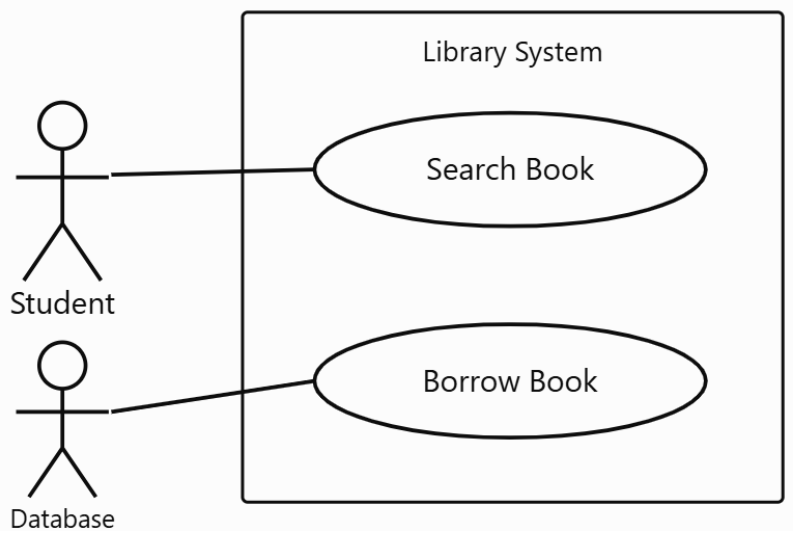


Figure 6. Incorrect diagram

Identify one mistake involving the actors.

Answer: The internal database should not be represented as an external actor. It is part of the library system, not a separate user or external system.

Correction: Remove the Database actor and its association. Connect the Student actor to both Search Book and Borrow Book use cases.

Q7. An online order has three states: Pending, Shipped, and Delivered. When an order is dispatched, its state changes from Pending to Shipped. When the customer receives it, the state changes to Delivered.

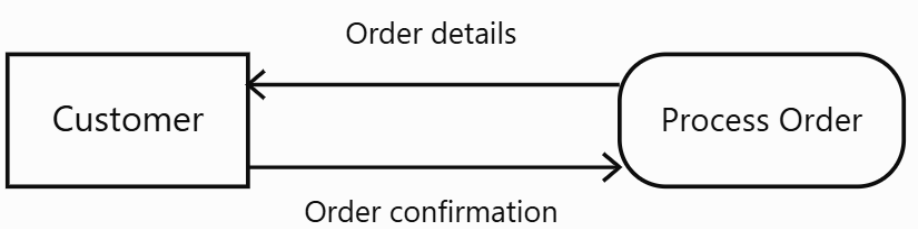


What is missing from this state diagram?

Answer: The transition from Shipped to Delivered is missing.

Correction: Add an arrow from Shipped to Delivered labelled order received.

Q8. An online shopping system receives order details from a customer. It processes the details and produces an order confirmation for the customer.



Identify the incorrect data flows.

Answer: Both data flow directions are reversed.

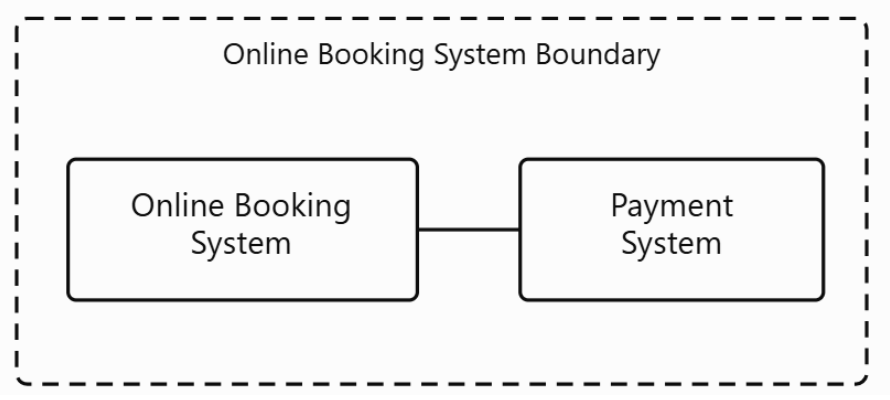
Correction:

Order details must flow from Customer to Process Order.

Order confirmation must flow from Process Order to Customer.

This is a useful reminder that a DFD shows how data moves between entities and processes, not just the order in which activities occur.

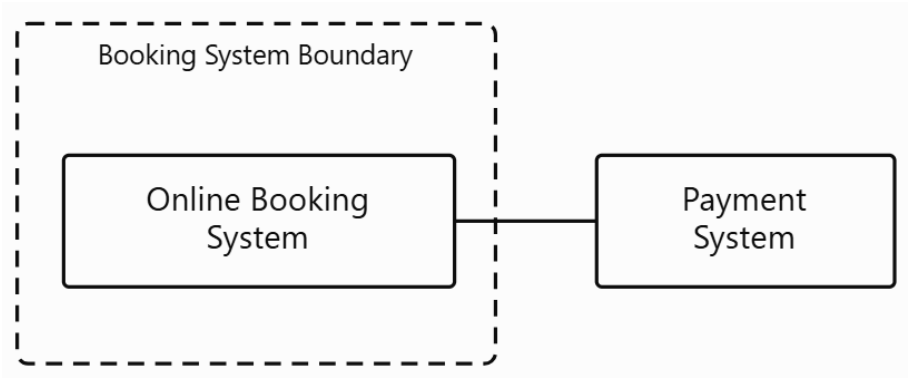
Q9. An online booking system communicates with an external payment system to process payments. The payment system is independently operated and is not part of the booking system.



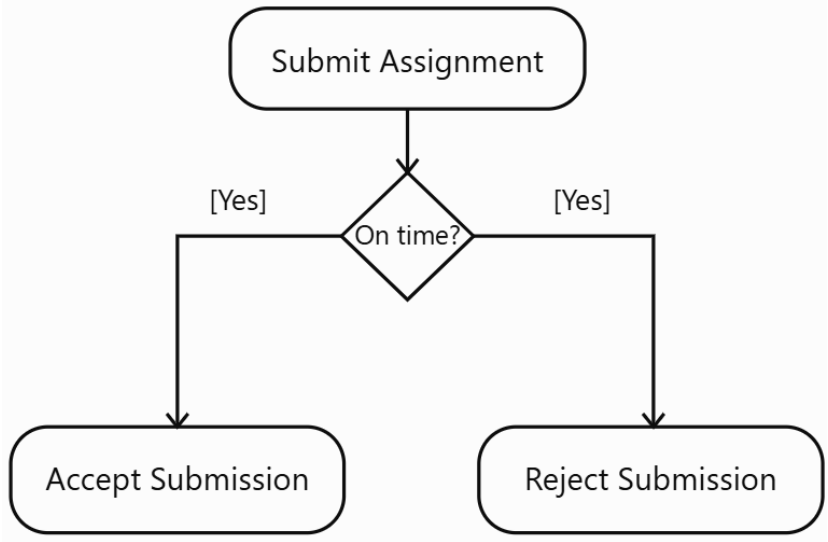
What is wrong with the system boundary?

Answer: The Payment System is incorrectly placed inside the Online Booking System boundary.

Correction: Move the Payment System outside the booking system boundary while keeping a connection between the two systems.



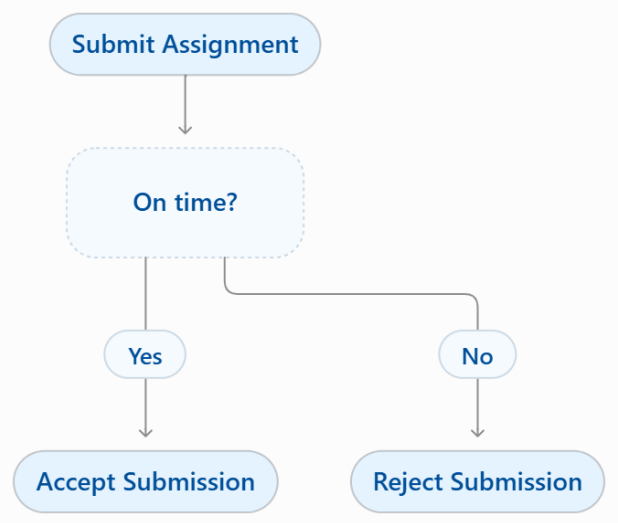
Q10. A student submits an assignment. The system checks whether the submission is on time. If on time, it accepts the submission. If late, it rejects the submission.



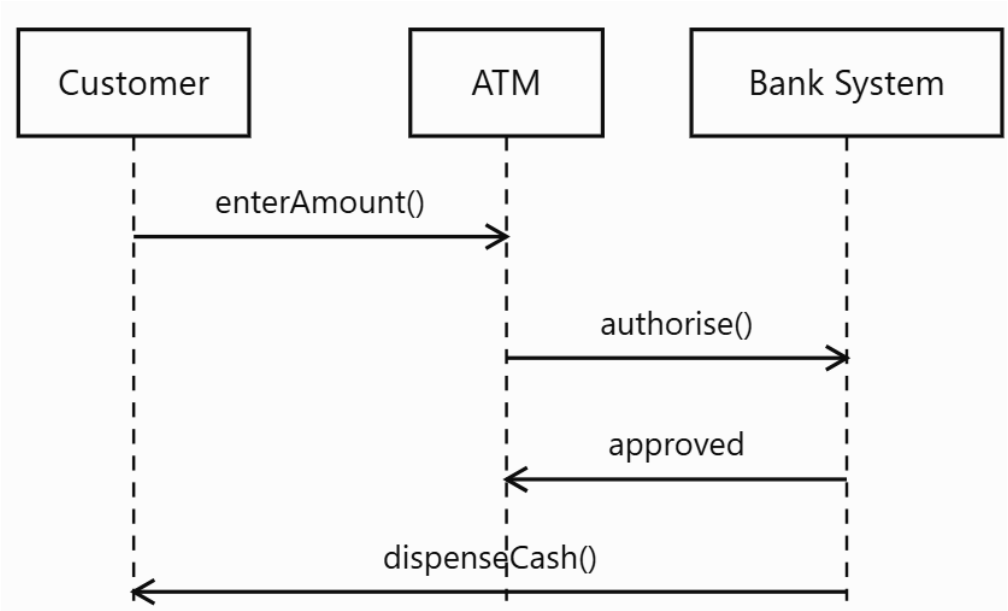
Identify the mistake in the decision conditions.

Answer: Both outgoing branches are labelled [Yes]. One of them must be [No].

Correction: The branch leading to Accept Submission should have [Yes], while the branch leading to Reject Submission should have [No].



Q11. An ATM system processes a cash withdrawal. The customer enters an amount, the ATM requests authorisation from the Bank System, and the ATM dispenses cash after receiving approval.



Which message is incorrectly represented?

Answer: The dispenseCash() message incorrectly originates from the Bank System and is sent to the Customer.

Correction: The ATM should perform the cash-dispensing operation after receiving authorisation from the Bank System.

